

TLEMCEN ZENATA, Algeria

WMO: 605310

Lat: 35.017N

Lon: 1.450W

Elev: 814

StdP: 14.27

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	35.8	38.4	28.4	23.1	50.2	31.6	26.8	50.3	27.4	58.4	23.6	57.3	1.6	120	0.397

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.5	97.7	70.4	94.4	70.0	91.4	69.9	76.2	88.5	74.8	86.5	73.6	84.8	10.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.1	126.6	81.3	71.6	120.3	79.7	70.1	114.2	78.8	40.1	88.7	38.8	86.4	37.6	84.8	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.9	17.6	15.4	DB	32.3	106.8	2.3	4.1	30.6	109.8	29.2	112.2	27.9	114.5	26.2	117.4	(4)
(5)			WB	30.8	78.4	2.6	1.8	29.0	79.7	27.4	80.8	26.0	81.8	24.1	83.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.1	52.3	53.4	57.0	60.6	66.7	73.6	79.3	80.3	74.4	68.5	59.5	54.1	(6)	
	DBStd	10.76	4.48	4.67	4.41	4.16	4.71	4.37	4.02	3.78	3.94	4.83	5.42	4.75	(7)	
	HDD50	65	27	18	4	0	0	0	0	0	0	0	2	14	(8)	
	HDD65	1683	395	325	249	142	34	0	0	0	0	19	181	338	(9)	
	CDD50	5564	97	113	222	319	519	709	910	939	733	574	286	143	(10)	
	CDD65	1709	0	1	2	11	88	259	445	474	284	128	16	1	(11)	
	CDH74	16625	3	11	65	191	855	2473	4731	5022	2195	942	127	10	(12)	
	CDH80	7210	0	3	12	39	250	959	2361	2500	781	278	25	2	(13)	
(14) Wind	WSAvg	5.0	5.3	5.3	4.9	5.2	5.3	5.3	5.3	5.0	4.5	4.2	5.0	4.8	(14)	
(15) Precipitation	PrecAvg	17.4	1.9	2.0	2.2	2.0	1.6	0.4	0.1	0.1	0.6	1.3	1.9	2.4	(15)	
	PrecMax	47.6	5.5	13.5	9.3	8.6	14.5	3.3	1.5	0.7	3.9	5.9	6.1	10.9	(16)	
	PrecMin	5.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)	
	PrecStd	9.1	1.4	2.0	2.0	1.9	2.4	0.7	0.2	0.2	0.7	1.4	1.3	2.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.0	74.1	80.5	84.6	91.5	97.1	103.5	104.1	95.1	92.9	83.8	74.4	(19)	
		MCWB	56.0	56.8	61.2	62.6	65.7	68.9	71.7	71.8	70.3	64.8	63.5	58.8	(20)	
	2%	DB	68.0	69.8	74.9	78.9	86.1	92.2	97.9	97.8	89.9	86.3	77.3	69.8	(21)	
		MCWB	55.0	56.4	59.4	61.3	64.8	68.8	70.5	71.1	69.0	65.5	60.4	57.0	(22)	
	5%	DB	64.6	66.3	71.4	75.2	82.3	88.2	94.4	94.0	87.4	82.5	73.4	66.5	(23)	
		MCWB	53.9	54.8	58.2	60.1	63.9	68.4	69.9	70.8	69.0	65.5	59.5	55.6	(24)	
	10%	DB	62.2	63.3	67.9	71.7	78.7	85.6	91.1	91.2	84.4	79.0	69.9	63.8	(25)	
		MCWB	53.1	54.2	57.2	59.1	63.0	67.6	69.9	71.2	68.8	65.0	58.9	54.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.9	60.4	64.6	66.3	70.2	74.3	77.7	78.6	75.2	71.0	66.7	61.7	(27)	
		MCDB	66.9	68.9	74.7	77.8	82.9	87.9	92.3	90.4	86.2	82.1	74.6	70.1	(28)	
	2%	WB	57.0	58.1	61.9	63.5	67.7	72.5	75.6	76.8	73.5	69.3	63.6	58.9	(29)	
		MCDB	64.3	65.9	71.5	73.9	79.8	86.1	89.3	88.9	83.7	79.6	72.7	66.1	(30)	
	5%	WB	55.3	56.4	59.7	61.8	66.0	70.9	74.3	75.5	72.2	67.9	61.8	57.1	(31)	
		MCDB	62.7	63.7	68.5	72.3	77.6	83.5	87.1	86.6	81.5	77.1	70.1	64.2	(32)	
	10%	WB	53.7	54.9	57.8	60.1	64.5	69.6	72.9	74.3	70.8	66.7	60.1	55.4	(33)	
		MCDB	60.6	61.9	65.7	69.4	75.2	81.0	84.5	84.9	79.8	75.5	67.8	62.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	19.1	19.5	21.0	21.7	22.0	22.5	23.0	22.5	21.2	21.5	19.3	18.8	(35)	
		MCDBR	24.0	24.8	27.8	28.9	29.2	27.1	28.5	27.3	25.0	26.5	24.7	22.7	(36)	
	5% WB	MCWBR	13.9	14.3	15.2	14.6	13.0	11.1	11.0	11.0	10.7	11.6	12.6	12.9	(37)	
		MCWBR	20.3	21.0	23.6	23.9	23.6	22.8	21.8	20.6	21.8	20.7	19.6	19.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.315	0.343	0.405	0.430	0.467	0.500	0.553	0.524	0.474	0.414	0.360	0.325	(40)	
		taud	2.507	2.426	2.225	2.169	2.090	2.002	1.871	1.952	2.092	2.255	2.410	2.499	(41)	
	Ebn at Noon		283	285	274	272	262	253	238	243	249	255	262	271	(42)	
		Edh at Noon	26	32	42	47	51	56	64	58	48	37	29	25	(43)	
(44) All-Sky Solar Radiation	RadAvg		902	1185	1579	1925	2199	2430	2396	2163	1758	1337	955	822	(44)	
	RadStd		80	101	104	138	155	78	61	66	75	86	78	57	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page